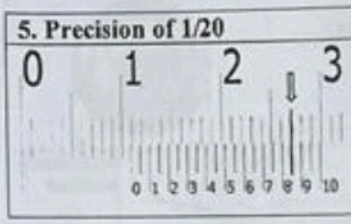
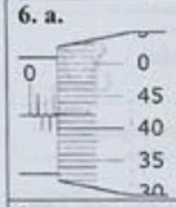
5. Precision of 1/20



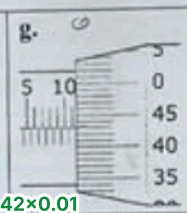
10 + 17×0.05 = 10.85 mm ✓

6. a.



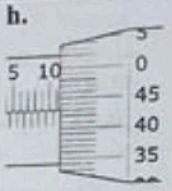
kovan 3 + tambur 42×0.01

g.



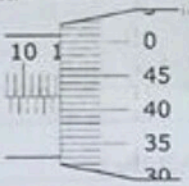
11 + 43×0.01

h.



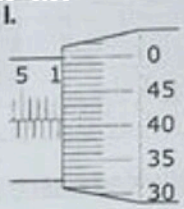
11 + 43×0.01

k.



14 + 42×0.01

l.



9.5 + 41×0.01

Q2. (P.Ç.10.2) Please answer questions in the test (15x2=30p.)

1. A part for a machine is to be made to the length specification 14.7 + 0.2mm. What is the largest acceptable size?

a) 14.5 mm ● 14.9 mm c) 16.7 mm d)14.8 mm

2. What is the point of having units in the sciences? → b

a) There really is no point; it's just a formality.
b) They distinguish our numbers as measurements instead of pure numbers.
c) They distinguish our numbers as pure numbers instead of measurements.
● We use units to tell us we are working with the metric system instead of the imperial system, which has no units.

3. A part for a machine is to be made to the length specification 17.8 + 0.4mm. The constructed part is 18.3mm. Is it acceptable?

● No b)Yes c) Not enough information

4) Which of the following represents an SI unit of luminous intensity? → b) Candela (cd)

a) Lumen ● b) Candela c) Dioptre ~~● None of them~~

5. What do we call an SI unit that is created by combining base units using algebra?

● a) a derived unit → a) derived unit
b) a divided unit
~~● a basic unit~~
d) a multiplied unit

9. In a measurement, what is the term used to specify the closeness of two or more measurements?

● Precision b) Accuracy c) Fidelity d) Threshold

10) When triangular distribution is used in a Type-B method?

a) When most values are near the origin
b) When most values are near the right most position of distribution
c) When most values are near the left most position of distribution
● When most values are near the centre of distribution

11. Closeness of measured value to true value is _____

● Accuracy b) Precision c) Correction d) Uncertainty

12) The mean length of an object is 5 cm. Which of the following measurements is most accurate?

● 4.9 cm (b) 4.805 cm (c) 5.25 cm (d) 5.4 cm

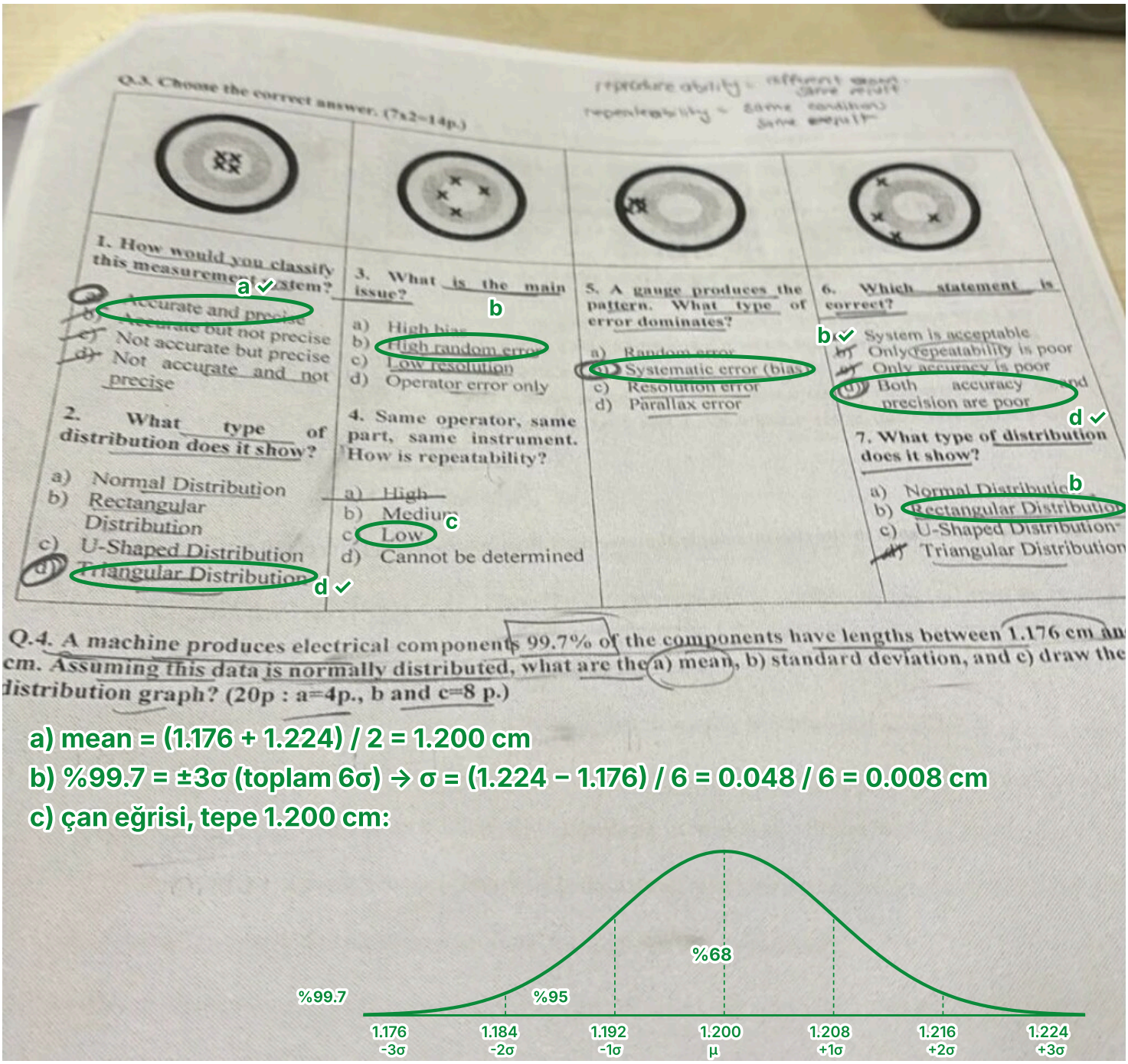
13. Which of the following statement is true about Type-A uncertainty?

Statement 1: Most of the systematic uncertainty is Type-A uncertainty.

Statement 2: Type-A uncertainties vary each and every time a measurement is made.

a) T, T ● b) F, T c) F, F ~~● T, F~~

→ b) F, T (Type A = random etki, sistematik değil)



Q.3. Choose the correct answer. (7x2=14p.)

1. How would you classify this measurement system?
a) Accurate and precise ✓ a
b) Accurate but not precise
c) Not accurate but precise
d) Not accurate and not precise

2. What type of distribution does it show?
a) Normal Distribution
b) Rectangular Distribution
c) U-Shaped Distribution
d) Triangular Distribution ✓ d

3. What is the main issue? b
a) High bias
b) High random error
c) Low resolution
d) Operator error only

4. Same operator, same part, same instrument. How is repeatability?
a) High
b) Medium
c) Low c
d) Cannot be determined

5. A gauge produces the pattern. What type of error dominates?
a) Random error
b) Systematic error (bias)
c) Resolution error
d) Parallax error

6. Which statement is correct? b ✓
a) System is acceptable
b) Only repeatability is poor
c) Only accuracy is poor
d) Both accuracy and precision are poor

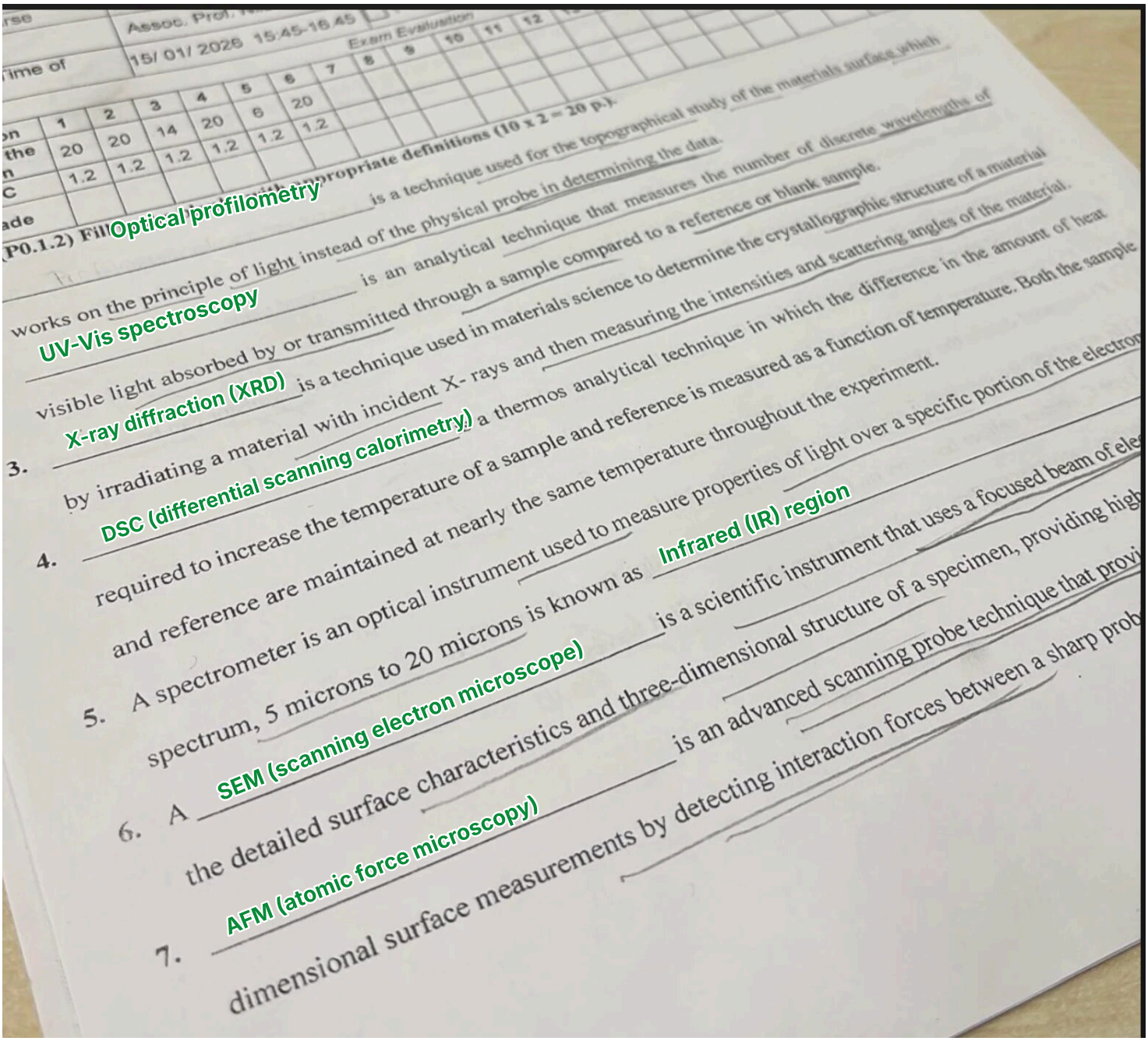
7. What type of distribution does it show? d ✓
a) Normal Distribution
b) Rectangular Distribution b
c) U-Shaped Distribution
d) Triangular Distribution

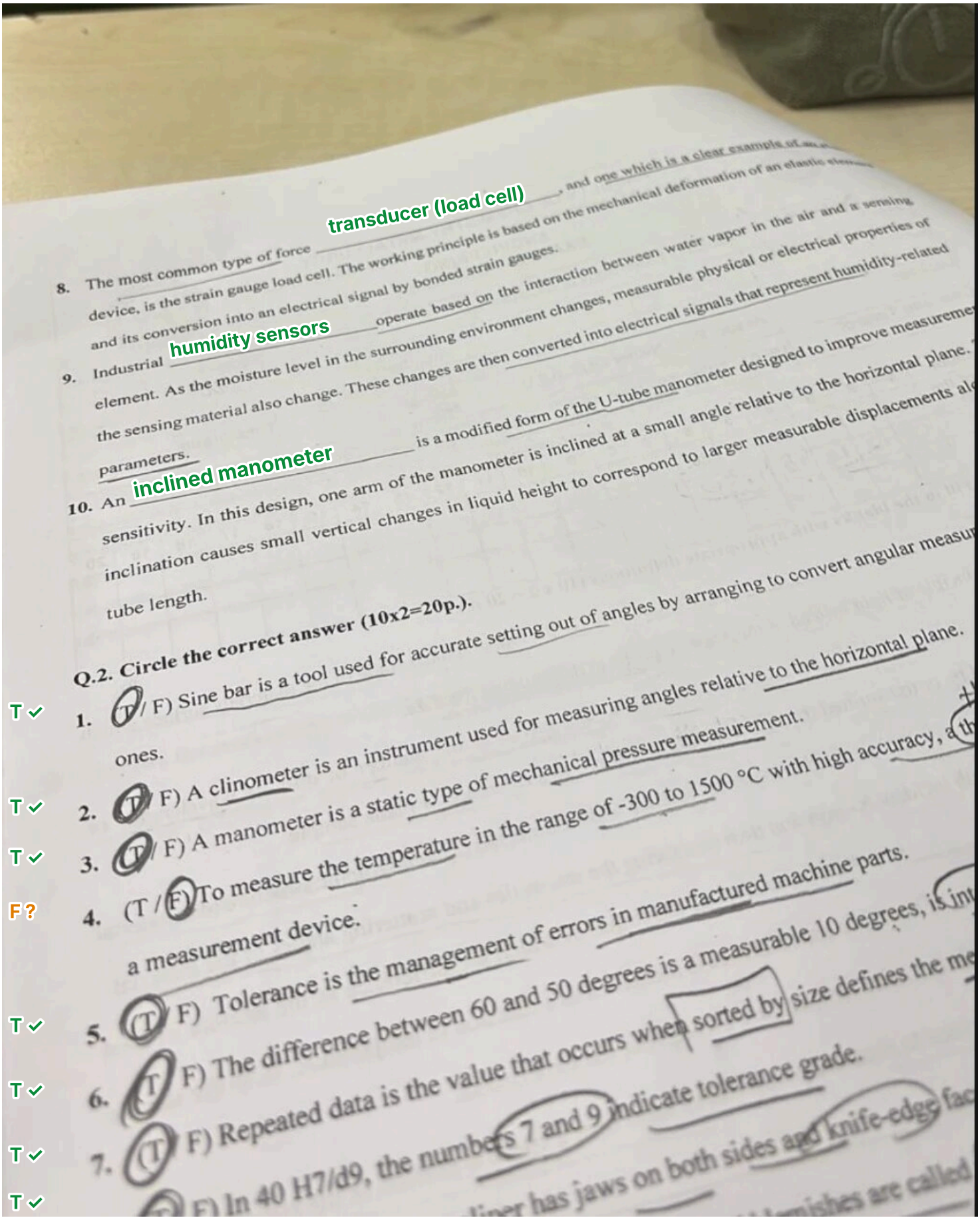
Q.4. A machine produces electrical components 99.7% of the components have lengths between 1.176 cm and cm. Assuming this data is normally distributed, what are the a) mean, b) standard deviation, and c) draw the distribution graph? (20p : a=4p., b and c=8 p.)

a) mean = (1.176 + 1.224) / 2 = 1.200 cm

b) %99.7 = ±3σ (toplam 6σ) → σ = (1.224 − 1.176) / 6 = 0.048 / 6 = 0.008 cm

c) çan eğrisi, tepe 1.200 cm:


Assoc. Prof.
15/ 01/ 2026 15:45-16:45
Exam Evaluation
1 2 3 4 5 6 7 8 9 10 11 12
20 20 14 20 6 20
1.2 1.2 1.2 1.2 1.2 1.2
(P0.1.2) Fill in the appropriate definitions (10 x 2 = 20 p.).
works on the principle of light instead of the physical probe in determining the data.
Optical profilometry
is a technique used for the topographical study of the materials surface which
visible light absorbed by or transmitted through a sample compared to a reference or blank sample.
UV-Vis spectroscopy
is an analytical technique that measures the number of discrete wavelengths of
3. by irradiating a material with incident X- rays and then measuring the intensities and scattering angles of the material
X-ray diffraction (XRD)
is a technique used in materials science to determine the crystallographic structure of a material
4. required to increase the temperature of a sample and reference is measured as a function of temperature. Both the sample
DSC (differential scanning calorimetry)
a thermos analytical technique in which the difference in the amount of heat
and reference are maintained at nearly the same temperature throughout the experiment.
5. A spectrometer is an optical instrument used to measure properties of light over a specific portion of the electro
spectrum, 5 microns to 20 microns is known as
Infrared (IR) region
6. A
SEM (scanning electron microscope)
is a scientific instrument that uses a focused beam of ele
the detailed surface characteristics and three-dimensional structure of a specimen, providing high
7.
AFM (atomic force microscopy)
is an advanced scanning probe technique that provi
dimensional surface measurements by detecting interaction forces between a sharp prob



8. The most common type of force transducer (load cell), and one which is a clear example of an
device, is the strain gauge load cell. The working principle is based on the mechanical deformation of an elastic
and its conversion into an electrical signal by bonded strain gauges.

9. Industrial humidity sensors operate based on the interaction between water vapor in the air and a sensing
element. As the moisture level in the surrounding environment changes, measurable physical or electrical properties of
the sensing material also change. These changes are then converted into electrical signals that represent humidity-related
parameters.

10. An inclined manometer is a modified form of the U-tube manometer designed to improve measureme
sensitivity. In this design, one arm of the manometer is inclined at a small angle relative to the horizontal plane.
inclination causes small vertical changes in liquid height to correspond to larger measurable displacements al
tube length.

Q.2. Circle the correct answer (10x2=20p.).

T ✓ 1. (T / F) Sine bar is a tool used for accurate setting out of angles by arranging to convert angular measu
ones.

T ✓ 2. (T / F) A clinometer is an instrument used for measuring angles relative to the horizontal plane.

T ✓ 3. (T / F) A manometer is a static type of mechanical pressure measurement.

F ? 4. (T / F) To measure the temperature in the range of -300 to 1500 °C with high accuracy, a th
a measurement device.

T ✓ 5. (T / F) Tolerance is the management of errors in manufactured machine parts.

T ✓ 6. (T / F) The difference between 60 and 50 degrees is a measurable 10 degrees, is int

T ✓ 7. (T / F) Repeated data is the value that occurs when sorted by size defines the me

T ✓ (T / F) In 40 H7/d9, the numbers 7 and 9 indicate tolerance grade.

T ✓ liner has jaws on both sides and knife-edge fac
nishes are called